VentureMiner AI

Memory Architecture

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Document 11 — Memory Architecture

Memory is what makes the system cumulative. It carries state across runs so the platform gets sharper with use. This document defines the four memory layers, their contracts, and their governance.

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1. Purpose & Scope

This document defines what the system remembers, for how long, and under what rules. It is the contract for cumulative learning without leaking across tenants.

2. Memory layers

Session	Single run	Run	In-process + scratchpad
Working	Single turn	User/turn	Redis (short TTL)
User	Per user account	User	Postgres + vector
Workspace	Per workspace	Workspace	Postgres + vector
Platform	All workspaces	Aggregate, anonymized	Postgres + vector

3. Session memory

- When it ends: when the run ends or is cancelled.
- Retention: none beyond the run.

4. Working memory

- What: the last N turns of a user session in the UI.
- Where: Redis, keyed by user.
- TTL: 1 hour idle, 8 hours absolute.
- Used for: multi-turn UI context, fast retrieval of recent items.

5. User memory

- What: durable user-level facts:
- Preferences (preferred rubric, watchlists).
- Calibration feedback (e.g. "I disagree with WTP for X — it should be lower").
- Decision history (which opportunities the user advanced, archived, or rejected).
- Where: Postgres memory.user_memory + vector embeddings.
- Used for: personalization; calibrating future runs.
- Retention: while the account is active; 30-day grace after deletion.

6. Workspace memory

- What: durable workspace-level facts:
- Common source priorities.
- Past reports and their acceptance feedback.
- Shared watchlists and rubrics.
- Where: Postgres memory.workspace_memory + vector.
- Used for: team-level personalization; cross-user learning.
- Retention: workspace lifetime; configurable per-tenant (Enterprise).

7. Platform memory

- What: aggregate, anonymized learnings:
- Which rubric dimensions correlate with accepted opportunities.
- Source reliability scores.
- Common claim patterns.
- Where: Postgres + vector, no tenant_id column.
- Used for: model selection, source prioritization, calibration.

8. Memory write discipline

- Atomic: writes are append-only or versioned; never destructive.
- Verified: the verifier audits memory writes for citation and policy.
- Versioned: every memory record has a version; supersession is explicit.
- Provenance: every record carries the source run_id and actor.

9. Memory retrieval

- Agents can memory.read(scope=user | workspace | platform, query=...) and get top-k records.
- Retrieval uses the same hybrid index as RAG (Document 10) but with a different namespace.

10. Privacy & isolation

- A user can only read their own user memory.
- A user can only read workspace memory for workspaces they belong to.
- No cross-workspace memory.
- Platform memory is anonymized; no per-tenant data appears in queries.

11. Retention & deletion

- User deletion cascades to user memory.
- Workspace deletion cascades to workspace memory.
- Platform memory is not affected by user/workspace deletion (anonymized).
- Right-to-be-forgotten: users can request deletion of all user memory; honored within 30 days.

12. Failure modes

Memory store down	Degrade: skip memory; run still works without personalization.
Corrupted record	Quarantine; alert; do not block run.
Provenance missing	Reject write.
Schema drift	Migration via Atlas; backwards-compatible.

13. Evaluation

- Calibration accuracy: does the system predict user acceptance better with memory? (Monthly)
- Memory precision: what fraction of recalled memories are useful? (Spot-checked)
- Cross-tenant leakage tests: monthly red-team.
- Acceptance rate of memory-informed reports vs. cold reports.

14.1 Glossary

Session memory	Transient state of a single run
Working memory	Recent turn context (Redis)
User memory	Per-user durable facts
Workspace memory	Per-workspace durable facts
Platform memory	Anonymized aggregate learnings

14.2 Revision history

v0.5	2026-07-20	Doc Team	All sections drafted
v1.0	2026-07-20	Doc Team	First approved version

14.3 Cross-references

- Multi-Agent: Document 08.
- RAG: Document 10.

End of Document 11 — Memory Architecture. The four-layer model is the contract for cumulative learning without cross-tenant leakage.